

Frobenius Algebras

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1 The Frobenius Theorem

Let A be a finite-dimensional K -algebra $a_1, a_2, \dots, a_n$ a basis of the K -vector space A , and $\alpha_{ijk} \in K$, $k \in \{1, 2, \dots, n\}$, the associated structure constants, that is,

$$a_j a_k = \sum_{i=1}^n \alpha_{ijk} a_i$$

for all $j, k \in \{1, 2, \dots, n\}$. Moreover, let $\lambda_j \in K$, $k \in \{1, 2, \dots, n\}$, be such that

$$1_A = \sum_{j=1}^n \lambda_j a_j$$

Invoking the structure constants α_{ijk} and the elements λ_j , which are equivalent to the associativity and identity conditions of the K -algebra. Moreover, we introduce the matrices

$$L(a_j) = [L(a_j)_{ik}] = [\alpha_{ijk}]_{ik} \in M_n(K) \quad , j \in \{1, 2, \dots, n\},$$

$$L(a_l) = [R(a_l)_{ik}] = [\alpha_{ilk}]_{ik} \in M_n(K) \quad , l \in \{1, 2, \dots, n\},$$

which determines K -linear maps

$$L : A \rightarrow M_n(K)$$

$$R^t : A \rightarrow M_n(K)$$

are representations of algebra A over K , called by Frobenius the first (left) regular representation and the second (right) regular representation of A over K , respectively. Frobenius also defined the matrices

$$P_h = [P_h]_{ij} = [\alpha_{ijk}]_{ij} \in M_n(K) \quad , h \in \{1, 2, \dots, n\}$$

and then the matrices

$$P(\xi) = \sum_{h=1}^n \xi_h P_h \quad , h \in \{1, 2, \dots, n\}$$

For all element $\xi = (\xi_1, \xi_2, \dots, \xi_n) \in K^n$, called by him *parastrophic matrices*, and showed they intertwine the left regular and the second regular representations of A .

Lemma 1. *For all elements $\xi \in K^n$ and $a \in A$ the following equality holds:*

$$P(\xi)L(a) = R^t(a)P(\xi)$$

The following theorem gives a characterization of finite dimensional K -algebras for which the first and second regular representations are equivalent.

Theorem 1.1. *Let A be a finite-dimensional K -algebra. Then the regular representations L and R^t of A are equivalent if and only if there is an invertible parastrophic matrix $P(\xi)$ for some $\xi \in K^n$*

In 1937-1939 R. Brauer, C. Nesbitt and T. Nakayama noticed the importance of the algebras discussed in the above Frobenius theorem, and proposed to call them Frobenius algebras.

Therefore, a finite dimensional K -algebra A over a field K is said to be a Frobenius Algebra if the first (left) regular representation L and the second (right) regular representation R^t of A over K equivalent for a chosen basis of A over K . Moreover, A is said to be symmetric algebra if there exists a symmetric invertible matrix P defining equivalence of L and R^t . Means, A is a Frobenius algebra (respectively, symmetric algebra) if and only if there exists an invertible (respectively, symmetric invertible) parastrophic matrix $P(\xi)$

2 The Brauer-Nesbitt-Nakayama Theorems

In This section, we state theorems that give a criteria for a finite-dimensional K -algebra A to be a Frobenius algebra (respectively, symmetric algebra), and are independent of the choice of a basis of A .

For a finite-dimensional K -algebra A over a field K , a K -bilinear form

$$(-, -) : A \times A \rightarrow K$$

is said to be associative if $(ab, c) = (a, bc)$ for all elements $a, b, c \in A$. Moreover, a K -bilinear form $(-, -) : A \times A \rightarrow K$ is said to be nondegenerate, if for every nonzero element $a \in A$, the linear forms $(a, -), (-, a) : A \rightarrow K$ are nonzero.

The following theorem is the first Brauser-Nesbitt-Nakayama theorem.

Theorem 2.1. *Let A be a finite-dimensional K -algebra over a field K . The following conditions are equivalent.*

1. *A is a Frobenius algebra.*
2. *There exists a nondegenerate associative K -bilinear form $(-, -) : A \times A \rightarrow K$*
3. *There exists a K -linear form $\phi : A \rightarrow K$ such that $\text{Ker} \phi$ does not contain a nonzero right ideal of A .*
4. *There exists an isomorphism $\theta : A_A \rightarrow D(A)_A$ of right A -modules.*

5. There exists a K -linear form $\phi' : A \rightarrow K$ such that $\text{Ker}\phi'$ does not contain a nonzero left ideal of A .
6. There exists an isomorphism $\theta' : {}_A A \rightarrow {}_A D(A)$ of left A -modules.

The second Brauer-Nesbitt-Nakayama theorem gives equivalent conditions for finite dimensional algebra to be a symmetric algebra.

Theorem 2.2. *Let A be a finite dimensional K -algebra over a field K . The following conditions are equivalent.*

1. A is a symmetric algebra.
2. There exists a nondegenerate associative symmetric K -bilinear form

$$(-, -) : A \times A \rightarrow K$$

3. There exists a K -linear form $\phi : A \rightarrow K$ such that $\phi(ab) = \phi(ba)$ for all $a, b \in A$, and $\text{Ker}\phi$ does not contain a nonzero one-sided ideal of A .
4. There exists an isomorphism $\theta : {}_A A_A \rightarrow {}_A D(A)_A$ of A -bimodules.

Corollary 2.3. *Let A be a finite-dimensional k -algebra. The following equivalences hold.*

1. A is a Frobenius algebra if and only if A^{op} is a Frobenius algebra.
2. A is a symmetric algebra if and only if A^{op} is a symmetric algebra.

Proposition 2.4. *Let $A_1, \dots, A_r$ be finite-dimensional K -algebras over a field K and $A = A_1 \times \dots \times A_r$. The following equivalences hold.*

1. A is a Frobenius algebra if and only if $A_1, \dots, A_r$ are Frobenius algebras.
2. A is a symmetric algebra if and only if $A_1, \dots, A_r$ are symmetric algebras.

3 Frobenius algebras

In this section, we present some properties of Frobenius algebras. We start with the following fact proved by T.Nakayama

Proposition 3.1. *Let A be a Frobenius K -algebra over a field K and $(-, -) : A \times A \rightarrow K$ be a nondegenerate associative K -bilinear form. Then there exists an automorphism ν of the K -algebra A such that*

$$(\nu(a), b) = (b, a)$$

for all elements $a, b \in A$.

For a Frobenius K -algebra A and a nondegenerate associative K -bilinear form $(-, -) : A \times A \rightarrow K$, a K -algebra automorphism $\nu : A \rightarrow A$ with $(\nu(a), b) = (b, a)$ for all $a, b \in A$ is said to be the Nakayama automorphism of A associated to $(-, -)$.

Lemma 2. *Let A be a Frobenius K -algebra over a field K , $(-, -) : A \times A \rightarrow K$ a nondegenerate associative K -bilinear form, ν the associated Nakayama automorphism of A , and c an invertible element of A . Then the map $\nu' : A \rightarrow A$, given by $\nu'(a) = \nu(cac^{-1})$ for all $a \in A$, is a Nakayama automorphism of A .*

The following proposition shows that a Nakayama automorphism of a Frobenius algebra A is uniquely determined up to an inner automorphism.

Proposition 3.2. *Let A be a Frobenius K -algebra over a field K , $(-, -)_1$ and $(-, -)_2$ two nondegenerate associative K -bilinear forms from $A \times A$ to K , and ν_1 and ν_2 the Nakayama automorphisms of A associated to these forms, respectively. Then there exists an invertible element $c \in A$ such that*

$$\nu_2(a) = \nu_1(cac^{-1}) \text{ for any } a \in A.$$

as a consequence of Prop. 3.2. we obtain the following characterization of the symmetric algebras in the class of Frobenius algebras.

Corollary 3.3. *Let A be a Frobenius K -algebra over a field K . The following conditions are equivalent.*

1. *A is a symmetric algebra.*
2. *Every Nakayama automorphisms of A is inner.*
3. *There exists an inner Nakayama automorphism of A .*

Let A be a finite-dimensional K -algebra and $D : \text{mod}A \rightarrow \text{mod}A^{\text{op}}$ the standard duality. Then D induces the dualities

$$\text{Proj}A \xrightleftharpoons[D]{D} \text{inj}A^{\text{op}}$$

$$\text{inj}A \xrightleftharpoons[D]{D} \text{Proj}A^{\text{op}}$$

between the categories of projective modules and injective modules in categories $\text{mod}A$ and $\text{mod}A^{\text{op}}$, respectively. As an immediate consequence, we have the following proposition.

Proposition 3.4. *Let A be a finite-dimensional K -algebra over a field K . The following conditions are equivalent.*

1. *A_A is an injective right A -module.*
2. *$\text{proj}A = \text{inj}A$*
3. *$\text{proj}A^{\text{op}} = \text{inj}A^{\text{op}}$*
4. *${}_A A$ is an injective left A -module.*

Recall that a finite-dimensional K -algebra with A_A is injective, equivalently ${}_A A$ injective, is called a selfinjective algebra. Therefore, a finite-dimensional K -algebra A is self-injective if the projective modules in $\text{mod}A$ coincide with the injective modules. In particular, A is a selfinjective algebra if and only if A^{op} is a selfinjective algebra.